

Global Bang Degree Does Not Control Sublevel Topology: A Level-Centered Repair

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Status. Substantial partial result and sharp obstruction, likely valid and pending human review. A fixed global Bang degree cannot bound the affine-section component constants of quasianalytic sublevel sets, even in a fixed analytic class on a fixed rectangle. Bang degrees of the shifted level functions $f - t$ and $f + t$ do give an explicit line-section bound.

1 Source prompt

For a sublevel set

$$K_t = \{x \in K : |f(x)| \leq t\},$$

Rainer proves a metric-entropy estimate containing a constant

$$C_{K_t} = C(n) \max_{1 \leq j \leq n-1} \omega_j B_{n-j}(K_t),$$

where $B_{n-j}(K_t)$ bounds the number of connected components of $P \cap K_t$ over affine $(n - j)$ -planes P . Definability gives finiteness, and the source asks for explicit bounds “via the Bang degree depending on the Denjoy–Carleman complexity of K_t ” [1, after Corollary 1.18].

The source does not define a numerical Denjoy–Carleman complexity or format for a definable presentation of K_t . The results below isolate a necessary scale that any such complexity must retain.

2 Proof intuition

Derivative bounds permit very high frequency when its amplitude is exponentially small. Adding that oscillation to a nonzero constant leaves the global size of the function, and therefore its global Bang degree, essentially unchanged. But taking a sublevel exactly at the constant term removes the large offset and exposes every oscillation topologically.

This also identifies the repair. On an interval, the boundary of $\{|f| \leq t\}$ consists of zeros of the two shifted functions $f - t$ and $f + t$. The source’s quasianalytic zero theorem applies to those shifts, so the relevant degrees must be centered at the boundary levels. Their norms detect the tiny oscillatory amplitude that the global degree misses.

3 Notation from the source

For an admissible weight $\mu = (\mu_j)_{j \geq 1}$, base derivative bound $M > 0$, length scale $a > 0$, and lower norm bound $b > 0$, write

$$\mathcal{N}(\mu, M, a, b) := 2D_{a\mu}(b/M)$$

$$B = \frac{C_N |K| \|g\|_K}{\nu_K \|\partial_{i_0}^2 f\|_K^{1/N}} + 2C N_{f,K} (|I|^{n-1} \|g\|_{I^n} + \|\partial_{i_0} g\|_{L^1(I^n)}),$$

$C > 0$ is an absolute constant, and $N_{f,K} \in \mathbb{N}$ is an integer depending only on f and K .

1.8. Metric entropy of sublevel sets. Let $\epsilon > 0$ and $K \subseteq \mathbb{R}^n$ relatively compact. The *covering number* $N_\epsilon(K)$ of K is the minimal number of closed balls of radius ϵ covering K and $H_\epsilon(K) := \log_2 N_\epsilon(K)$ is the ϵ -*entropy* of K .

Corollary 1.18. *In the setup of Theorem 1.1, assume additionally that $K \in \mathcal{K}(\mathbb{R}^n)$ is definable in $\mathbb{R}_{C_M}$ and $L = B(0, r)$. Then the covering number of the sublevel set K_t of $f|_K$ satisfies*

$$N_\epsilon(K_t) \leq C(n) \frac{|K|}{\nu_K} \left(\frac{C_N t}{\|f\|_K} \right)^{1/N} \frac{1}{\epsilon^n} + C_{K_t} \left(1 + \left(\frac{r}{\epsilon} \right)^{n-1} \right), \quad t > 0,$$

for a constant $C_{K_t} > 0$.

Proof. This follows from Corollary 1.6 and [YC04, Proposition 5.6]. □

The constant C_{K_t} is of the form

$$C_{K_t} = C(n) \max_{1 \leq j \leq n-1} \omega_j \cdot B_{n-j}(K_t),$$

where $B_{n-j}(K_t)$ bounds the number of connected components of $P \cap K_t$, where P is any affine $(n-j)$ -plane in $\mathbb{R}^n$. The bounds $B_{n-j}(K_t)$ exist because K_t is definable in $\mathbb{R}_{C_M}$. It would be interesting to find explicit bounds via the Bang degree depending on the Denjoy–Carleman complexity of K_t .

1.9. Organization of the paper. In Section 2, we introduce the admissible weights that control the derivatives of quasianalytic functions and define the Bang degree. We also introduce terminology that accounts for the fact that often differentiability of finite order is sufficient.

In Section 3, we recall basics on o-minimal structures, in particular, the expansion of the real field by restricted function in a quasianalytic Denjoy–Carleman class.

In Section 4, we formulate the general version of the quasianalytic Remez inequality (Theorem 4.13) and discuss the collection $\mathcal{K}(\mathbb{R}^d)$ of sets that support this inequality. Its proof comprises Sections 5 and 6.

In Section 7, we deduce estimates for the volume of sublevel sets and obtain a

Figure 1: The metric-entropy corollary and the prompt on source PDF page 9.

for the associated degree of the source, whenever the quadruple is admissible. For quasianalytic μ it is always finite. Proposition 10.1 of the source states that a (μ, M) -smooth function on an interval of length a whose supremum norm is at least b has at most $\mathcal{N}(\mu, M, a, b)$ zeros, counted with multiplicity [1, Proposition 10.1].

4 A fixed-class obstruction

Theorem 1 (global degree cannot control topology). *Let*

$$K = [0, 2\pi] \times [0, 1], \quad \mu_j = j, \quad M_0 = 2.$$

For every integer $N \geq 1$, there is a restricted analytic (μ, M_0) -smooth function $f_N : K \rightarrow \mathbb{R}$ such that the associated global degree with lower bound $b_0 = 1$ is independent of N , while

$$B_1(\{x \in K : |f_N(x)| \leq 1\}) \geq N + 1.$$

In particular, no bound depending only on the global Bang degree, the fixed weight and derivative majorant, the dimension, and the fixed geometry of K can control the component constants in the source prompt.

Proof. Set

$$f_N(x, y) = 1 + e^{-N} \sin(Nx).$$

For $j \geq 1$, only the pure x -derivative contributes to the source's multivariate derivative seminorm, and

$$\|f_N\|_{j,K} = e^{-N} N^j \leq j! \leq 2j!. \quad (1)$$

Indeed, $j! \geq (j/e)^j$, and with $u = j/N$ the desired first inequality follows from

$$\left(\frac{j}{eN}\right)^j \geq e^{-N} \iff N(u \log u - u + 1) \geq 0.$$

Also $\|f_N\|_K = 1 + e^{-N} < 2$. Since the weight $\mu_j = j$ gives $M_j = 2j!$, equation (1) proves $(\mu, 2)$ -smoothness for every N . The fixed lower bound $\|f_N\|_K \geq 1$ therefore uses the same associated degree

$$\mathcal{N}(\mu, 2, \text{diam } K, 1)$$

for the whole family. The degree obtained by inserting the actual norm $\|f_N\|_K$ is no larger, by monotonicity in the lower bound.

Because $1 - e^{-N} > 0$, the level-one sublevel condition is

$$|f_N(x, y)| \leq 1 \iff \sin(Nx) \leq 0.$$

On $[0, 2\pi]$ this set is the isolated point $\{0\}$ together with the N disjoint intervals

$$\left[\frac{(2k+1)\pi}{N}, \frac{(2k+2)\pi}{N}\right], \quad k = 0, \dots, N-1.$$

Their products with $[0, 1]$ are $N+1$ connected components. Intersecting with any horizontal line gives the same $N+1$ components, proving the asserted lower bound for B_1 . $\square$

5 The level-centered repair

Theorem 2 (explicit line-section bound). *Let $I \subset \mathbb{R}$ be a nondegenerate compact interval, let μ be a quasianalytic admissible weight, and let $g : I \rightarrow \mathbb{R}$ be (μ, M_0) -smooth. Fix $t > 0$ and put*

$$\widehat{M}_0 = 2(M_0 + t), \quad b_\sigma = \|g - \sigma t\|_I \quad (\sigma \in \{-1, 1\}).$$

If g is not identically t or $-t$, then

$$\text{comp}\{s \in I : |g(s)| \leq t\} \leq 1 + \sum_{\sigma \in \{-1, 1\}} \mathcal{N}(\mu, \widehat{M}_0, |I|, b_\sigma). \quad (2)$$

If $g \equiv t$ or $g \equiv -t$, the sublevel set is I and has one component.

Proof. Assume g is not identically either boundary level. The boundary in I of $S = \{|g| \leq t\}$ is contained in

$$\{g - t = 0\} \cup \{g + t = 0\}.$$

Consequently, if $Z_I(h)$ denotes the number of distinct zeros of h in I ,

$$\text{comp}(S) \leq Z_I(g - t) + Z_I(g + t) + 1. \quad (3)$$

Both shifted functions are $(\mu, \widehat{M}_0)$ -smooth: their zeroth norms are at most $M_0 + t < \widehat{M}_0$, and all positive-order derivatives are those of g . Moreover $b_\sigma > 0$. The source’s zero theorem bounds $Z_I(g - \sigma t)$ by $\mathcal{N}(\mu, \widehat{M}_0, |I|, b_\sigma)$. Substitution in (3) proves (2). $\square$

Corollary 1 (uniform planar entropy constant). *Let $K \subset \mathbb{R}^2$ be a convex body and let f be (μ, M_0) -smooth on a convex neighborhood of K . Fix $t > 0$. Suppose there is $b_* > 0$ such that for every affine line P , every nondegenerate interval $I = P \cap K$, and both $\sigma \in \{-1, 1\}$, either $f|_I \equiv \sigma t$ or*

$$\|f|_I - \sigma t\|_I \geq b_*.$$

Then, with

$$\mathcal{N}_* = \mathcal{N}(\mu, 2(M_0 + t), \text{diam } K, b_*),$$

one has

$$B_1(K_t) \leq 2\mathcal{N}_* + 1.$$

Thus the component term in the constant of Corollary 1.18 of the source may be taken at most $C(2)\omega_1(2\mathcal{N}_ + 1)$.*

Proof. Every nonempty line section of a convex body is an interval. Parameterizing it by arclength preserves the derivative majorant. Apply Theorem 2; use $|I| \leq \text{diam } K$, the assumed lower bound, and monotonicity of $\mathcal{N}$ in its length and lower-bound arguments. A restriction identically equal to a boundary level contributes one component and obeys the same final bound. $\square$

6 Scope, verification, and novelty audit

The construction and estimates are exact and have no computational dependency. The elementary inequality in (1) is uniform in both j and N . The isolated endpoint component at $x = 0$ is included, so the count is $N + 1$, not merely N . In the same family, $\|f_N - 1\|_K = e^{-N}$; hence the level-centered associated degree is at least twice its starting index, which grows linearly with N . It therefore records exactly the scale erased by the constant offset.

This does not solve the source’s full higher-dimensional request. The phrase “Denjoy–Carleman complexity” is not assigned a format or numerical value in the paper, and scalar line restrictions do not bound the number of components of higher-dimensional affine sections. Such a result would require effective control of multivariate critical systems or an effective stratification theorem for a specified Denjoy–Carleman presentation.

A bounded novelty audit checked the run’s result, attempt, and proof-gap indexes, arXiv:2606.08183, the exact metric-entropy prompt, and the zero-count result cited in the source. No prior packet or later resolution was found. The source is very recent and a final external-source refresh was unavailable, so novelty remains pending specialist review.

Recommended human-review focus is: (1) whether the intended notion of Denjoy–Carleman format was meant to include a separate frequency-sensitive presentation parameter; (2) the fixed-weight derivative estimate; and (3) the passage from the two shifted zero counts to the planar uniform constant.

References

- [1] A. Rainer, *Effective quasianalytic Remez inequalities on tame sets*, arXiv:2606.08183 (2026).